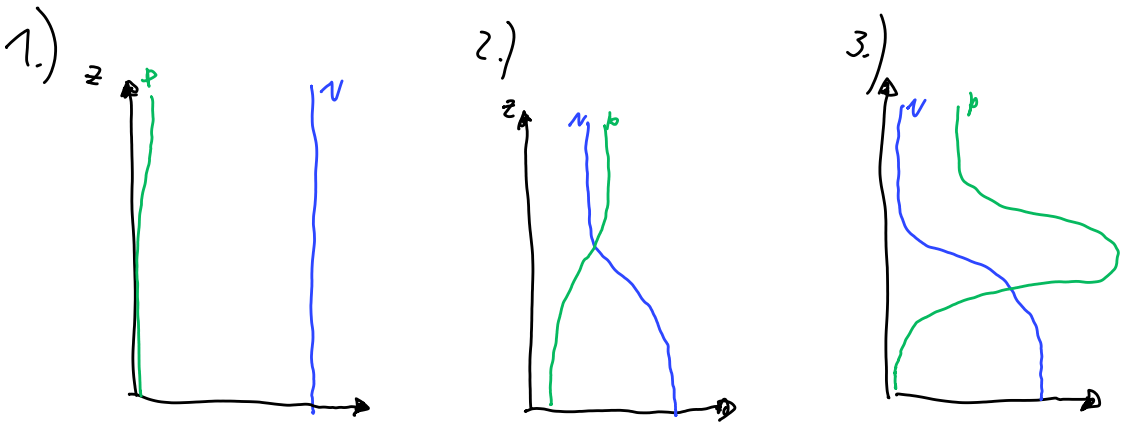
Computational Marine Ecological Modelling, Exercise 3

The aim of this exercise is to combine the NPZD models and the water column model.

1. Make an water-column NPZD model with constant light. Use a diffusive boundary condition at the bottom for nutrients. Run it to convergence. Plot vertical profiles of: light, nutrients etc. at the final time step.
2. Make a plot which illustrates the limiting resource for phytoplankton (light or nutrients) as a function of depth.

Optional

3. Run the model with a seasonally varying light.
4. Experiment with various model formulations, e.g.:
   a. different version of the functional response
   b. with different boundary conditions on the bottom
   c. Different closure mortality

$$L(z) = L_0 e^{-k_w t} \qquad k_w = \frac{1}{L}$$